CanPlan

Reflection

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1. Introduction

During this internship at CanAssist, I worked on improving and expanding the CanPlan application. CanPlan is an application designed to support people who benefit from step-by-step guidance during daily activities.

The internship focused on several different aspects of software development, including redesigning and rebranding the application, improving accessibility, researching cloud-based expansion possibilities, implementing new features, and exploring future backend architectures.

Besides technical development, the internship also included research, documentation, user feedback analysis, testing, and collaboration with supervisors and designers.

This reflection document discusses both the substantive outcomes of the internship project and my personal learning experiences throughout the internship period.

2. Substantive reflection

2.1. Accomplishments

During the internship, significant improvements were made to the CanPlan application both visually and technically. One of the largest accomplishments was the complete redesign and rebranding of the application. The user interface was modernized with a stronger focus on accessibility, consistency, and cognitive simplicity.

Several new concepts and features were also explored and implemented, including a branching task system, a simplified interface mode, reminder functionality, and improvements to task workflows.

Besides frontend improvements, extensive research was conducted regarding the future expansion of CanPlan into a cloud-supported platform. Different backend technologies, databases, hosting solutions, and security considerations were analyzed to determine which technologies would best support long-term scalability and maintainability.

In addition, concepts and designs for a future admin dashboard were created to support remote task management and user administration.

The internship therefore contributed both practical improvements to the existing application and technical research that can support future development within CanAssist.

2.2. Remaining Future Work

Although many improvements and research activities were completed during the internship, several parts of the project remain future work. The cloud-based architecture and backend infrastructure were mainly explored conceptually and were not yet fully implemented in production.

Future work could include implementing full synchronization between devices, adding authentication systems, deploying the backend infrastructure, and further expanding the administrative dashboard.

Additional user testing with larger groups of users could also provide valuable feedback regarding accessibility and usability improvements.

Because CanPlan continues to evolve, the project created a strong foundation for future development rather than representing a completely finalized platform.

2.3. Current State of the Project

Several improvements created during the internship have already contributed to the ongoing development of CanPlan. The redesign work and accessibility improvements helped modernize the application and provided clearer design direction for future updates.

In addition, several newly implemented features have already been integrated into the current version of the application. Features such as the branching task system, simplified user workflows, accessibility-focused interface improvements, and the implementation of a simple mode already contribute to a more intuitive and flexible user experience. Improvements to task organization and navigation also help make the application easier to use for people who benefit from structured step-by-step guidance.

The research regarding cloud architecture, backend technologies, deployment strategies, and future synchronization also created valuable technical documentation that can support future development decisions within the organization.

While some concepts remain in the research or prototype phase, such as the cloud infrastructure and administrative dashboard, the internship work has already influenced future planning and development discussions surrounding CanPlan.

2.4. Advice for the Future

For the future, I would recommend continuing to prioritize accessibility and cognitive simplicity throughout the development of CanPlan. Since the application supports users with different needs, maintaining a simple and intuitive user experience remains extremely important.

I would also recommend preserving the strong offline functionality of the application while gradually expanding cloud-based features in a controlled and modular way.

In addition, more user testing and feedback sessions could help further improve usability and identify additional accessibility improvements.

From a technical perspective, maintaining a modular architecture and scalable backend structure will help simplify future expansion and long-term maintenance of the platform.

3. Personal reflection

3.1. Personal Growth

This internship allowed me to experience what it is like to work within a professional environment on a long-term software project. Throughout the internship, I became more independent in planning my work, conducting research, and communicating technical ideas and progress with supervisors and team members.

I also learned how important communication and feedback are during software development projects. Regular discussions with supervisors and designers helped me better understand the importance of collaboration and iterative improvement.

The internship also improved my ability to manage larger projects over a longer period of time, balancing implementation work, research, testing, and documentation simultaneously.

Besides the professional side of the internship, living and working abroad had a big impact on me personally. I had to take care of everything myself, from planning my days to managing transport and daily responsibilities, which made me much more independent. I also had the chance to travel and explore the area, which helped me see new places and experience a different environment and culture. Overall, this period abroad was a unique experience that I will not forget easily and it helped me grow a lot in independence and self-organization.

3.2. Technical Growth

From a technical perspective, the internship allowed me to deepen my knowledge in mobile development and broaden my understanding of supporting backend and hosting technologies. A significant part of my work was focused on Flutter, where I further improved my ability to develop and structure cross-platform applications for both iOS and Android

A major learning experience was the iOS build and deployment process. I worked with Xcode to generate and troubleshoot iOS builds, and I became more familiar with signing configurations, provisioning profiles, and build-related issues that are specific to Apple's ecosystem. In addition, I learned how to manage releases through App Store Connect, including uploading builds and preparing them for testing and distribution via TestFlight.

Next to mobile development, I conducted extensive research into backend and hosting solutions to better understand how the application could be supported in a scalable and maintainable way. This included evaluating different architectural approaches, deployment options, and hosting strategies, and understanding the trade-offs between them in terms of complexity, scalability, and long-term maintainability.

This combination of practical mobile development work and research into supporting infrastructure helped me develop a more complete technical perspective on how mobile applications are built, deployed, and maintained in a production environment.

3.3. Challenges During the Internship

One of the largest challenges during the internship was the implementation of the branching task system. During development, it became clear that the original data structure was not suitable for supporting more advanced task flows. As a result, part of the implementation needed to be redesigned and rebuilt. Although this initially felt frustrating, it taught me the importance of scalable architecture and careful planning.

Another challenge involved researching future cloud-based expansion possibilities. Because many different technologies and architectural approaches were available, it was sometimes difficult to determine which solutions would best fit the long-term goals of CanPlan.

Balancing accessibility and functionality was also challenging. Since the application is intended for users with different cognitive needs, it was important to simplify the interface while still preserving useful functionality.

Finally, technical limitations related to iOS deployment and outdated hardware introduced additional complications during testing and deployment preparation.

3.4. Final Reflection

Overall, this internship was a highly valuable learning experience that allowed me to grow both technically and personally. I not only improved my programming and software architecture knowledge, but also developed stronger communication, research, and project management skills.

Working on a project focused on accessibility and user support also gave me a better understanding of how technology can positively impact people's daily lives.

The internship provided me with valuable real-world experience and gave me a much clearer understanding of large-scale software projects, collaborative development, and future-oriented system design.